

Henrique Lopes | Curriculum Vitae

Fafe, Portugal

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in henrique-lopes-643357316 • Born on October 30, 1999

Summary

Bachelor in Engineering Physics and Master's in Interactive Technologies and Game Design. Experience in research and development of Artificial Intelligence software, digital interface design, accessibility, web development, and virtual reality. Focused on UX/UI, prototyping, and integration of AI in interactive systems.

Education

Faculty of Engineering, University of Porto (FEUP)

Master in Multimedia – Interactive Technologies and Digital Games

Porto

2023–2025

University of Minho

Bachelor in Engineering Physics

Braga

2017–2023

Professional Experience

INESC TEC

Research Intern

Porto

2024–2025

- Development of Virtual Reality applications integrating conversational AI (GPT).
- Collaboration with multidisciplinary teams (developers, designers, and researchers).
- Creation of functional prototypes with Unity, APIs, and integration layers.
- Conducting usability tests in immersive environments.

Academic and Research Projects

Virtual Moderator in VR: Master's thesis in collaboration with INESC TEC: VR environment for social interaction with GPT.

ThinkAlike: Collaborative platform with AI-assisted UI.

StoryShow: Interactive website with secure PHP backend.

Physics Game: Educational game in C simulating forces and collisions.

Technical Skills

Programming: JavaScript (ES6+), TypeScript, C, C++, C#, Haskell, Assembly, Python.

Web Development: React 18, React Router v6, React Hooks, React Hook Form, Node.js, Express, Flask, Django, FastAPI, PHP.

Frontend and UI/UX: HTML, CSS, Tailwind CSS, CSS Modules, p5.js, Figma, prototyping and accessibility (WCAG).

Databases: SQL (PostgreSQL, MySQL), NoSQL (MongoDB, Redis), Mongoose, SQLAlchemy ORM.

APIs and Integrations: REST, GraphQL, Axios, Fetch API, i18n/React i18next, JWT, CORS, Async/await and Promises, HTTP, TLS.

Artificial Intelligence: LangChain, Chroma, Llama, GPT, Deepseek, integration of AI into applications.

DevOps and Cloud: Docker, Docker Compose, Kubernetes, AWS (CI/CD), Azure, Ansible, Nginx, Terraform.

Tools and Monitoring: Git, GitHub, VS Code, Postman, Swagger, Flasgger, Prometheus, Grafana, Loki, Sentry, ESLint, Jest, Jasmine.

Collaboration and Management: Scrum, Kanban, Agile methodologies.

Other: Unity, Simulink, LabView, Reaper, Pure Data.

Languages

Portuguese: Native

English: Advanced